Carbon-Taxed Transformers: Efficient, Accurate, and Sustainable LLMs

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Abstract

The growing adoption of Large Language Models in software engineering has introduced unsustainable computational and environmental costs. We propose Carbon-Taxed Transformers (CTT), a compression pipeline inspired by economic carbon taxation, integrating Neural Architecture Search, structured pruning, quantization, and knowledge distillation in a principled sequence. CTT generalizes across encoder-only, encoder-decoder, and decoder-only architectures, and is evaluated on code clone detection, summarization, and generation. Results show up to 49× memory reduction, 10× speedup, and an 81% reduction in CO_2 emissions, while retaining 98% accuracy in clone detection, 89% in summarization, and up to 91% in generation. Ablation studies confirm both the pipeline ordering and each component are essential.

CCS Concepts

• **Computing methodologies** → *Artificial intelligence*; **Machine learning approaches**; • **Software and its engineering** → *Software maintenance tools.*

Keywords

Model Compression, Green Computing, Energy-Efficient Inference

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1 Introduction

Large Language Models (LLMs) have transformed software engineering (SE), enabling advances in different SE tasks [6, 15]. However, their deployment carries significant computational and environmental costs: GPT-3 alone needs 325 GB of GPU memory [10]. Critically, inference accounts for over 90% of total ML energy consumption [7], and a single LLM can emit tons of CO_2 [13].

Model compression techniques such as knowledge distillation (KD) [9], quantization [4], and pruning [5], offer promising directions. However, prior SE works [8, 11, 12] typically apply a single technique to one architecture, leaving combined, multi-architectural compression unexplored. While PEFT methods such as LoRA [2] reduce fine-tuning costs, they retain the full model during inference and thus do not address deployment efficiency.

We propose **Carbon-Taxed Transformers (CTT)**, a compression pipeline that operationalizes a computational *carbon tax*, algorithmic constraints forcing principled trade-offs between performance and efficiency. In this work, we: (1) propose CTT with *architectural generalization* across encoder-only, decoder-only, and encoder-decoder models, (2) introduce *principled pipeline ordering* (NAS→pruning→quantization→KD), validated via ablation studies, (3) enforce *budget-aware optimization* with hard constraints on latency, memory, and CO_2, and (4) demonstrate up to 49× memory reduction, 10× speedup, and 81% CO_2 reduction while retaining 89–98% task performance across three core SE tasks. Further details, including experimental setup and comprehensive analysis, are available in the full paper [1].

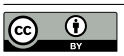
2 Methodology

This section summarizes the methodology from our full paper [1]. CTT targets four key architectural parameters: hidden layers, attention heads, hidden dimensions, and feedforward dimension size (Figure 1 adapted from our full paper [1]).

Stage 1: Structural Reduction. We apply NAS over a predefined configuration space to identify an initial compact architecture, then refine it through structured pruning while monitoring performance. An empirical analysis confirmed that deeper layers often act as near-identity mappings with minimal representational gain, providing principled cut-off points.

Stage 2: Quantization. The pruned model undergoes quantization *before* training, imposing a "tax on precision" which helps to adapt to quantization-induced noise during inference, mitigating the accuracy degradation observed in post-training quantization [4].

Stage 3: KD. The quantized student is trained to match the teacher's soft output distribution via KD loss, acting as a "performance rebate" that recovers accuracy lost during the compression.



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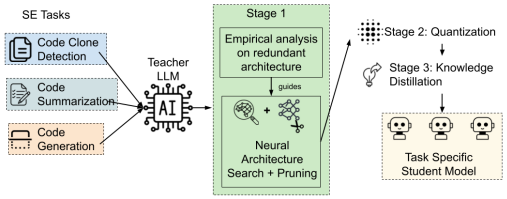

Figure 1: CTT pipeline (adapted from our full paper [1]).

Table 1: Inference-time CO_2 emissions.

Task	Model	Size	CO_2 (kg)	Δ
CCD	UniXCoder (Teacher)	504 MB	0.00421	–
	CTT Student	**1.6 MB**	**0.00198**	↓53%
Summ.	CodeT5-base (Teacher)	892 MB	0.00936	–
	CTT Student	**65 MB**	**0.00549**	↓41%
Gen.	CodeGen-350M (Teacher)	1.4 GB	0.03455	–
	CTT Student	**52.5 MB**	**0.00669**	↓81%
	Qwen 1.5B (Teacher)	3.1 GB	0.04300	–
	CTT Student	**216 MB**	**0.01318**	↓69%

3 Results and Discussion

We evaluate CTT on code clone detection (CCD) using BigCloneBench [14] and UniXCoder as teacher (encoder-only), code summarization using CodeSearchNet summarization dataset [3] and CodeT5-base [15] (encoder-decoder), and code generation using CodeSearchNet (code generation)+MBPP and CodeGen-350M and Qwen Coder 2.5 1.5B (decoder-only). All results and discussion reported in this section are summarized from our full paper [1].

Performance. The CTT student retains 98% of teacher performance in CCD while being 300× smaller. It also outperforms CodeBERT and CodeT5 in recall while matching CodeT5+ and Compressor [12], and outperforms Avatar [11] in precision. For summarization, it preserves 89% ROUGE-L at 92.7% size reduction, outperforming CodeT5-small across BLEU, METEOR, and ROUGE-L. In generation, the CodeGen-derived student maintains 91% BLEU at 97% reduction; the Qwen-derived student retains 80% BLEU at 93% reduction, with pass@1 reaching 68% of the teacher's.

Efficiency. CTT achieves up to 8× GPU and 10× CPU inference speedups in CCD, 3× in summarization, and 3–6× in generation. Memory reductions reach up to 49× on GPU and 8× on CPU.

Environmental Impact. Table 1 summarizes inference-time CO_2 emissions from our full paper [1], measured using CodeCarbon. CTT students consistently emit substantially less CO_2.

Ablation Insights. Two ablation studies validate CTT's design. First, our NAS→pruning→quantization ordering outperforms all permutations as alternative orderings increase latency by 4–6% and CO_2 by up to 25%. Second, removing KD causes accuracy to collapse to ∼52% (near-random), confirming that compression delivers efficiency while KD restores task-specific knowledge. Across all tasks, the average teacher–student gap is +2.1 points (95% bootstrap CI [+0.0, +4.3]; paired t-test $p = 0.23$), showing no statistically significant degradation. CTT's one-time training overhead is offset by inference savings that dominate long-term energy costs [7].

4 Conclusion

CTT is a compression pipeline integrating NAS, pruning, quantization, and KD that yields compact, fast, environmentally aligned SE models: up to 49× memory reduction, 10× speedup, and 81% CO_2 reduction while retaining 89–98% task performance. Its composable ordering and budgeting give developers fast on-device tools, enterprises low-cost deployment, and researchers compact models under modest resources.

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